

Math Midterm Review — Algebra, Quadratics and Statistics

One note for the whole midterm. Built from Session M1 (exponents to quadratic factorization), Session M2 (data and statistics), the Statistics Test Review and the Hypothesis Testing note, plus a new section on quadratic equations (section 4).

How to use it (45 minutes): 10 min sections 1–3 (recall) · 20 min section 4 (quadratics, new) · 10 min section 5 (statistics) · 5 min section 6 (mixed check).

1. Laws of exponents

For $a, b \neq 0$:

Law	Rule	Example
Product	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^4 = 2^7 = 128$
Quotient	$a^m \div a^n = a^{m-n}$	$5^6 \div 5^4 = 5^2 = 25$
Power of a power	$(a^m)^n = a^{mn}$	$(3^2)^3 = 3^6 = 729$
Power of a product	$(ab)^m = a^m b^m$	$(2x)^3 = 8x^3$
Power of a quotient	$(\frac{a}{b})^m = \frac{a^m}{b^m}$	$(\frac{2}{3})^2 = \frac{4}{9}$
Zero exponent	$a^0 = 1$	$7^0 = 1$
Negative exponent	$a^{-n} = \frac{1}{a^n}$	$2^{-3} = \frac{1}{8}$
Rational exponent	$a^{m/n} = (\sqrt[n]{a})^m$	$8^{2/3} = (\sqrt[3]{8})^2 = 4$

Why $a^0 = 1$: $a^n \div a^n = a^{n-n} = a^0$, and any nonzero number divided by itself is 1.

Same base, equal powers: if $a^x = a^y$ (with $a > 0, a \neq 1$), then $x = y$. Example:
 $2^{x+1} = 32 = 2^5 \Rightarrow x = 4$.

Common mistakes: $(a+b)^2 \neq a^2 + b^2 \cdot -3^2 = -9$ but $(-3)^2 = 9 \cdot 2x^{-1} = \frac{2}{x}$, not $\frac{1}{2x}$.

2. Polynomials, identities and the two theorems

Degree = highest power of the variable. Linear (1), quadratic (2), cubic (3).

Identities (memorize):

$$\begin{aligned}(a+b)^2 &= a^2 + 2ab + b^2 & (a-b)^2 &= a^2 - 2ab + b^2 & a^2 - b^2 &= (a+b)(a-b) \\(x+a)(x+b) &= x^2 + (a+b)x + ab \\a^3 + b^3 &= (a+b)(a^2 - ab + b^2) & a^3 - b^3 &= (a-b)(a^2 + ab + b^2)\end{aligned}$$

Remainder Theorem. When a polynomial $p(x)$ is divided by $(x-a)$, the remainder is $p(a)$.

Factor Theorem. $(x-a)$ is a factor of $p(x)$ exactly when $p(a) = 0$.

Example. Is $(x-2)$ a factor of $p(x) = x^3 - 3x^2 + 4$? $p(2) = 8 - 12 + 4 = 0$, so yes.

3. Factoring — pick the method in this order

1. **Common factor first:** $6x^2 + 9x = 3x(2x + 3)$.
2. **Two terms:** difference of squares $a^2 - b^2 = (a + b)(a - b)$. Example:
 $4x^2 - 25 = (2x + 5)(2x - 5)$. A sum of squares $a^2 + b^2$ does not factor over the real numbers.
3. **Three terms** $ax^2 + bx + c$: split the middle term. Find two numbers with **product** ac and **sum** b , rewrite bx with them, then factor by grouping.
4. **Four terms:** group in pairs.

Why splitting works: $(px + q)(rx + s) = prx^2 + (ps + qr)x + qs$. The two middle pieces ps and qr add to b and multiply to $(pr)(qs) = ac$.

Example ($a = 1$). $x^2 + 7x + 12$: product 12, sum 7 $\rightarrow$ 3 and 4. Answer $(x + 3)(x + 4)$.

Example ($a \neq 1$). $6x^2 + 11x - 10$: product -60 , sum 11 $\rightarrow$ 15 and -4 .

$$6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5) = (3x - 2)(2x + 5)$$

Sign guide: $c > 0 \rightarrow$ both numbers have the sign of b . $c < 0 \rightarrow$ opposite signs; the larger one has the sign of b .

Always check by multiplying back.

4. Quadratic equations (new)

Standard form: $ax^2 + bx + c = 0$, with $a \neq 0$. The solutions are also called **roots** or **zeros**; on a graph they are the **x-intercepts**.

4.1 Four ways to solve

(1) Factoring + Zero Product Property. If $A \cdot B = 0$ then $A = 0$ or $B = 0$. One side must be 0 first.

$$x^2 - 5x = 14 \Rightarrow x^2 - 5x - 14 = 0 \Rightarrow (x - 7)(x + 2) = 0 \Rightarrow x = 7 \text{ or } x = -2$$

(2) Square roots (when there is no x term). Remember the $\pm$.

$$3x^2 - 48 = 0 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4 \qquad (x - 1)^2 = 9 \Rightarrow x - 1 = \pm 3 \Rightarrow x = 4 \text{ or } -2$$

(3) Completing the square. Add $\left(\frac{b}{2}\right)^2$ to both sides (when $a = 1$).

$$x^2 + 6x - 7 = 0 \Rightarrow x^2 + 6x = 7 \Rightarrow x^2 + 6x + 9 = 16 \Rightarrow (x + 3)^2 = 16 \Rightarrow x = -3 \pm 4$$

So $x = 1$ or $x = -7$.

(4) Quadratic formula. Works for every quadratic.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

4.2 Derivation of the quadratic formula (completing the square on the general equation)

$$ax^2 + bx + c = 0$$

Divide by a and move the constant:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Add $\left(\frac{b}{2a}\right)^2$ to both sides:

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Take square roots and solve for x :

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example. $2x^2 - 3x - 5 = 0$: $a = 2$, $b = -3$, $c = -5$.

$$b^2 - 4ac = 9 + 40 = 49 \quad x = \frac{3 \pm 7}{4} \Rightarrow x = \frac{5}{2} \text{ or } x = -1$$

4.3 The discriminant $D = b^2 - 4ac$

Discriminant	Roots	Graph
$D > 0$	two different real roots (rational if D is a perfect square)	crosses the x-axis twice
$D = 0$	one repeated real root, $x = -\frac{b}{2a}$	touches the x-axis at the vertex
$D < 0$	no real roots (two complex roots)	never meets the x-axis

Example. $x^2 + 4x + 5 = 0$: $D = 16 - 20 = -4 < 0$, no real roots. With $i = \sqrt{-1}$ (so $i^2 = -1$):
 $x = \frac{-4 \pm 2i}{2} = -2 \pm i$.

4.4 Sum and product of the roots

If the roots are r_1 and r_2 :

$$r_1 + r_2 = -\frac{b}{a} \quad r_1 r_2 = \frac{c}{a}$$

Why: expand $a(x - r_1)(x - r_2)$ to get $ax^2 - a(r_1 + r_2)x + ar_1 r_2$, then match the coefficients with $ax^2 + bx + c$.

A quadratic with roots r_1, r_2 : $x^2 - (\text{sum})x + (\text{product}) = 0$. Roots 3 and -5 : $x^2 + 2x - 15 = 0$.

4.5 Graph of $y = ax^2 + bx + c$ (a parabola)

Feature	How to find it
Direction	$a > 0$ opens up (minimum) · $a < 0$ opens down (maximum)
Axis of symmetry	$x = -\frac{b}{2a}$
Vertex	$x = -\frac{b}{2a}$, then substitute to get y
y-intercept	$(0, c)$
x-intercepts	solve $ax^2 + bx + c = 0$

Three forms:

- Standard: $y = ax^2 + bx + c$ (shows the y-intercept c)
- Vertex: $y = a(x - h)^2 + k$ (vertex (h, k))
- Factored (intercept): $y = a(x - p)(x - q)$ (x-intercepts p and q ; axis $x = \frac{p+q}{2}$)

Example. $y = x^2 - 4x - 5$. Axis $x = \frac{4}{2} = 2$. Vertex $y = 4 - 8 - 5 = -9$, so vertex $(2, -9)$, a minimum. Vertex form $y = (x - 2)^2 - 9$. Factored $y = (x - 5)(x + 1)$, x-intercepts 5 and -1 . y-intercept $(0, -5)$.

4.6 Word problem pattern (US units)

Height of an object thrown upward: $h(t) = -16t^2 + v_0t + h_0$ (feet, seconds).

Example. A ball is thrown up at 48 ft/s from a height of 64 ft: $h = -16t^2 + 48t + 64$.

- **Maximum height:** $t = -\frac{48}{2(-16)} = 1.5$ s; $h = -16(1.5)^2 + 48(1.5) + 64 = 100$ ft.
- **Hits the ground:** $-16t^2 + 48t + 64 = 0$; divide by -16 : $t^2 - 3t - 4 = 0$; factor: $(t - 4)(t + 1) = 0$; so $t = 4$ s (reject $t = -1$).

4.7 Hidden quadratics (from M1)

Substitute to make it a quadratic. $4^x - 6 \cdot 2^x + 8 = 0$: let $u = 2^x$, so $u^2 - 6u + 8 = 0$, $(u - 2)(u - 4) = 0$, $2^x = 2$ or 4 , so $x = 1$ or $x = 2$.

5. Statistics

5.1 Collecting data

Term	Meaning
Population / sample	the whole group / the part actually measured
Parameter / statistic	a number describing a population (μ, σ) / describing a sample ($\bar{x}$)
Bias	a method that systematically favors some outcomes

Sampling methods: simple random (every member equally likely) · systematic (every k th) · stratified (random sample from every group) · cluster (all members of a few random groups) · convenience and self-selected (both biased).

Ways to collect data: survey · observational study (watch, no treatment; shows association only) · experiment (treatment is assigned at random; the only method that can show cause) · simulation.

Bias to look for: undercoverage, non-response, leading question wording, self-selection.

5.2 Center

- **Mean:** $\bar{x} = \frac{\sum x}{n}$. Grouped data: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, where x_i is the class mark (midpoint).
- **Median:** middle value of the ordered data (average of the two middle values when n is even). Not pulled by outliers.
- **Mode:** most frequent value.
- Grouped median: $l + \frac{\frac{n}{2} - cf}{f} \times h$ · Grouped mode: $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$ (l = lower limit of the class, h = class width, cf = cumulative frequency before the median class, f_1 = modal frequency, f_0 and f_2 = the frequencies before and after).
- Skewed data: report the median. Symmetric data: mean = median.

5.3 Spread

- **Range** = max – min.

- **Standard deviation:** $\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$; variance = σ^2 .

Check. Scores 6, 7, 7, 8, 8, 8, 8, 9, 9, 10: mean 8; squared distances add to 12; $\sigma = \sqrt{1.2} \approx 1.10$.

5.4 Graphs

Bar graph: categories, gaps between bars. **Histogram:** continuous classes, no gaps; with unequal class widths plot adjusted frequency = $\frac{\text{frequency}}{\text{class width}} \times \text{smallest width}$. Frequency polygon: join the class marks.

5.5 Normal distribution

Bell-shaped, symmetric about the mean.

- **Empirical rule:** 68% within $\mu \pm \sigma$ · 95% within $\mu \pm 2\sigma$ · 99.7% within $\mu \pm 3\sigma$.
- **z-score:** $z = \frac{x - \mu}{\sigma}$; back to a value: $x = \mu + z\sigma$.
- The table gives the area to the **left** of z . Right: $1 - \text{table}$. Between: $\text{table}(b) - \text{table}(a)$.

Example. $\mu = 70$, $\sigma = 5$. What percent is between 65 and 80? 65 is -1σ , 80 is $+2\sigma$:

$34\% + 34\% + 13.5\% = 81.5\%$. And $x = 82$ gives $z = \frac{82 - 70}{5} = 2.4$.

5.6 Margin of error and hypothesis testing

- **Margin of error** for a sample proportion: $\pm \frac{1}{\sqrt{n}}$. For $n = 400$: $\pm 0.05 = \pm 5\%$. A poll result of 52% means the true value is likely between 47% and 57%. To halve the margin of error, use 4 times the sample size.
- **Hypothesis test.** Null hypothesis = the claim being tested. Significance level α = the cutoff (usually 5%).

Rule: if the observed result (or one more extreme) shows up in α or less of the simulated samples, **reject** the null hypothesis. Otherwise **do not reject** (this does not prove the claim).

Example. 36 heads in 50 flips appeared in 0 of 200 fair-coin simulations ($0\% \leq 5\%$): reject "the coin is fair". 29 heads appeared in 39 of 200 ($19.5\% > 5\%$): do not reject.

6. Mixed check (answers below)

1. Simplify $\frac{(2x^3)^2 \cdot x^{-4}}{4x}$.
2. Evaluate $27^{2/3}$.
3. Factor $3x^2 - 12$.
4. Factor $2x^2 + 7x + 3$.
5. Solve $x^2 - 9x + 20 = 0$.
6. Solve $x^2 + 4x - 1 = 0$ (exact form).
7. How many real roots does $3x^2 - 2x + 5 = 0$ have?
8. Find the vertex of $y = -2x^2 + 8x - 3$. Is it a maximum or a minimum?
9. Is $(x + 1)$ a factor of $x^3 + 2x^2 - x - 2$?
10. Test scores are normal with $\mu = 500$, $\sigma = 100$. Find the z-score of 650 and the percent of scores between 400 and 600.
11. A poll of 900 people finds 60% in favor. Give the margin of error and the interval.
12. A result occurred in 8 of 200 simulations. At $\alpha = 0.05$, reject or not?

Answers, step by step

1. $(2x^3)^2 = 4x^6$; $4x^6 \cdot x^{-4} = 4x^2$; $\frac{4x^2}{4x} = x$.
 2. $\sqrt[3]{27} = 3$; $3^2 = 9$.
 3. $3(x^2 - 4) = 3(x + 2)(x - 2)$.
 4. Product 6, sum 7 $\rightarrow$ 6 and 1: $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3)$
 $= (2x + 1)(x + 3)$.
 5. $(x - 4)(x - 5) = 0$, so $x = 4$ or $x = 5$.
 6. $x = \frac{-4 \pm \sqrt{16 + 4}}{2} = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$.
 7. $D = 4 - 60 = -56 < 0$: no real roots.
 8. $x = -\frac{8}{2(-2)} = 2$; $y = -8 + 16 - 3 = 5$. Vertex $(2, 5)$, a maximum because $a < 0$.
 9. $p(-1) = -1 + 2 + 1 - 2 = 0$: yes.
 10. $z = \frac{650 - 500}{100} = 1.5$. 400 to 600 is $\mu \pm \sigma$: 68%.
 11. $\frac{1}{\sqrt{900}} = \frac{1}{30} \approx 0.033$, so about $\pm 3.3\%$: 56.7% to 63.3%.
 12. $\frac{8}{200} = 4\% \leq 5\%$: reject the null hypothesis.
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Notes for next session

- Quadratics (section 4) is new: check completing the square and the discriminant first.
- Confirm with the school's topic list which statistics parts are on the midterm.